

[MS-RAI]:

Remote Assistance Initiation Protocol

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1 Introduction

The Remote Assistance Initiation Protocol is a set of **Distributed Component Object Model (DCOM)** interfaces, as specified in [\[MS-DCOM\]](#), for initiating a **Remote Assistance connection** to another computer in a **domain**. The Remote Assistance Initiation Protocol allows an authorized **expert** to start **Remote Assistance** on a remote **novice** computer to retrieve data that is required to make a Remote Assistance connection from the expert computer to the novice computer.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

class identifier (CLSID): A GUID that identifies a software component; for instance, a DCOM object class or a COM class.

computer name: The DNS or NetBIOS name.

Distributed Component Object Model (DCOM): The Microsoft Component Object Model (COM) specification that defines how components communicate over networks, as specified in [\[MS-DCOM\]](#).

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a domain controller (DC) and host a member list that identifies all members of the domain, as well as optionally hosting the Active Directory service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

domain name: A domain name or a NetBIOS name that identifies a **domain**.

expert: The side of a **Remote Assistance connection** that is able to view the remote screen of the other computer in order to provide help.

fully qualified domain name (FQDN): An unambiguous domain name that gives an absolute location in the Domain Name System's (DNS) hierarchy tree, as defined in [\[RFC1035\]](#) section 3.1 and [\[RFC2181\]](#) section 11.

novice: The side of a **Remote Assistance connection** that shares its screen with the other computer in order to receive help.

opnum: An operation number or numeric identifier that is used to identify a specific **remote procedure call (RPC)** method or a method in an interface. For more information, see [\[C706\]](#) section 12.5.2.12 or [\[MS-RPCE\]](#).

Remote Assistance (RA): A feature of the operating system that allows screen, keyboard, and mouse sharing so that a computer user can be assisted by a remote helper.

Remote Assistance connection: A communication framework that is established between two computers that facilitates **Remote Assistance**.

Remote Desktop Protocol (RDP): A multi-channel protocol that allows a user to connect to a computer running Microsoft Terminal Services (TS). RDP enables the exchange of client and server settings and also enables negotiation of common settings to use for the duration of the connection, so that input, graphics, and other data can be exchanged and processed between client and server.

remote procedure call (RPC): A communication protocol used primarily between client and server. The term has three definitions that are often used interchangeably: a runtime environment providing for communication facilities between computers (the RPC runtime); a set of request-and-response message exchanges between computers (the RPC exchange); and the single message from an RPC exchange (the RPC message). For more information, see [C706].

terminal services (TS): A service on a server computer that allows delivery of applications, or the desktop itself, to various computing devices. When a user runs an application on a terminal server, the application execution takes place on the server computer and only keyboard, mouse, and display information is transmitted over the network. Each user sees only his or her individual session, which is managed transparently by the server operating system and is independent of any other client session.

Unicode string: A Unicode 8-bit string is an ordered sequence of 8-bit units, a Unicode 16-bit string is an ordered sequence of 16-bit code units, and a Unicode 32-bit string is an ordered sequence of 32-bit code units. In some cases, it could be acceptable not to terminate with a terminating null character. Unless otherwise specified, all **Unicode strings** follow the UTF-16LE encoding scheme with no Byte Order Mark (BOM).

universally unique identifier (UUID): A 128-bit value. UUIDs can be used for multiple purposes, from tagging objects with an extremely short lifetime, to reliably identifying very persistent objects in cross-process communication such as client and server interfaces, manager entry-point vectors, and **RPC** objects. UUIDs are highly likely to be unique. UUIDs are also known as globally unique identifiers (GUIDs) and these terms are used interchangeably in the Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the UUID. Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [C706] has to be used for generating the UUID.

well-known endpoint: A preassigned, network-specific, stable address for a particular client/server instance. For more information, see [C706].

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[C706] The Open Group, "DCE 1.1: Remote Procedure Call", C706, August 1997, <https://publications.opengroup.org/c706>

Note Registration is required to download the document.

[MS-DCOM] Microsoft Corporation, "[Distributed Component Object Model \(DCOM\) Remote Protocol](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-ERREF] Microsoft Corporation, "[Windows Error Codes](#)".

[MS-OAUT] Microsoft Corporation, "[OLE Automation Protocol](#)".

[MS-RA] Microsoft Corporation, "[Remote Assistance Protocol](#)".

[MS-RDPBCGR] Microsoft Corporation, "[Remote Desktop Protocol: Basic Connectivity and Graphics Remoting](#)".

[MS-RPCE] Microsoft Corporation, "[Remote Procedure Call Protocol Extensions](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

1.2.2 Informative References

[MSDN-CRYPTO] Microsoft Corporation, "Cryptography Reference", <http://msdn.microsoft.com/en-us/library/aa380256.aspx>

1.3 Overview

The Remote Assistance Initiation Protocol provides a set of **DCOM** interfaces that enable an **expert** to retrieve the **Remote Assistance** connection-specific data from the remote **novice** computer. This Remote Assistance connection-specific data is subsequently used to initiate a **Remote Assistance connection** as explained in the Remote Assistance Initiation Protocol.

The expert needs to have the IP address or **fully qualified domain name (FQDN)** of the novice computer to use this protocol.

The expert is the DCOM client and the novice is the DCOM server.

Before the expert's DCOM call is executed on the novice computer, DCOM performs a check to verify that the expert is on the list of authorized Remote Assistance helpers on the novice computer. [<1>](#)

1.4 Relationship to Other Protocols

The Remote Assistance Initiation Protocol relies on the OLE Automation Protocol [\[MS-OAUT\]](#), the **Distributed Component Object Model (DCOM)** Remote Protocol [\[MS-DCOM\]](#), and on the Microsoft **remote procedure call (RPC)**, as specified in the Remote Procedure Call Protocol Extensions [\[MS-RPCE\]](#).

The Remote Assistance Protocol [\[MS-RA\]](#) is dependent on both the Remote Assistance Initiation Protocol and the Remote Desktop Protocol: Basic Connectivity and Graphics Remoting [\[MS-RDPBCGR\]](#).

The following diagram illustrates the relationships between the preceding protocols.

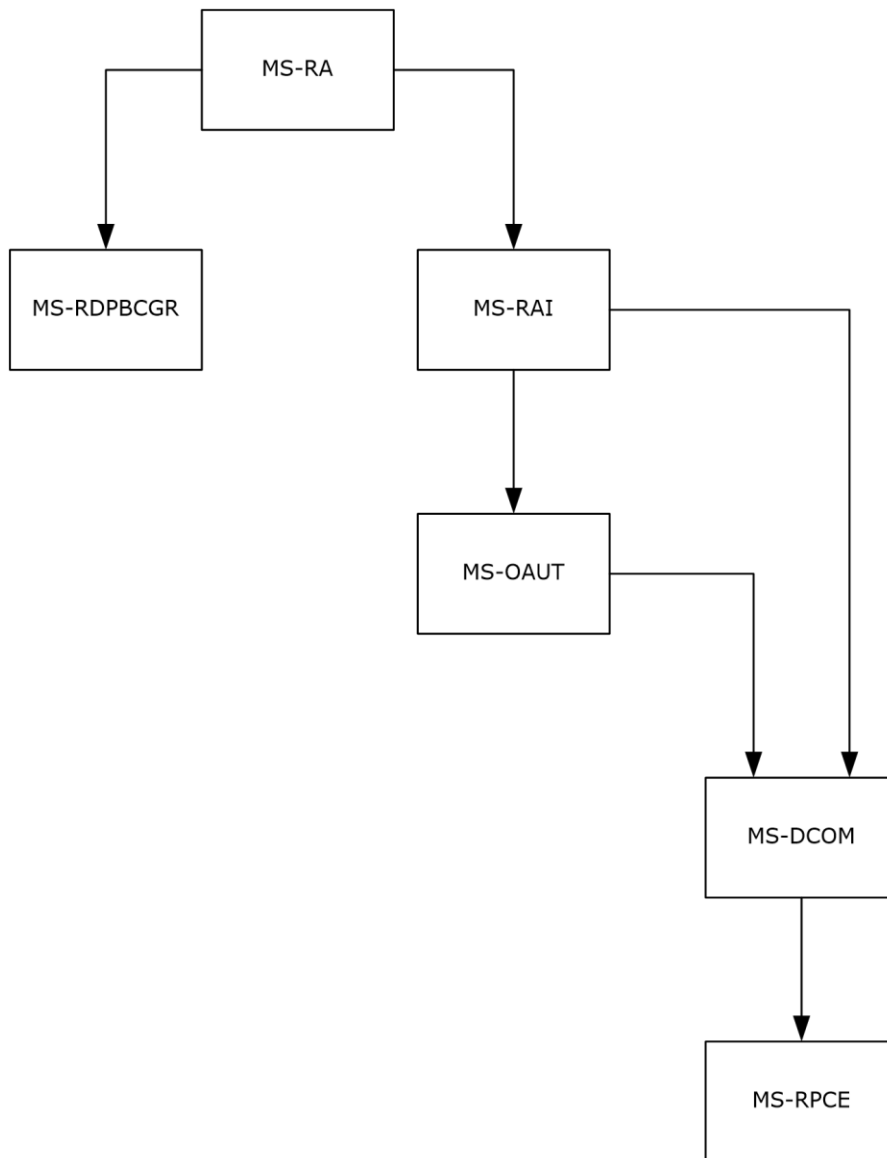


Figure 1: Relationships between protocols

1.5 Prerequisites/Preconditions

This protocol is implemented over **DCOM** and **RPC**, and, as a result, has the prerequisites specified in the Distributed Component Object Model (DCOM) Remote Protocol [\[MS-DCOM\]](#) and Remote Procedure Call Protocol Extensions [\[MS-RPCE\]](#) as being common to DCOM and RPC interfaces.

The Remote Assistance Initiation Protocol assumes that the **expert** has the IP Address or the **FQDN** of the **novice** computer.

1.6 Applicability Statement

This protocol is used to perform the following functions:

- Using the **novice** IP address or **FQDN**, the **expert** queries the novice for a list of active **terminal services** sessions running on the novice computer. Additional information about each terminal services session—DomainName, UserName, terminal services SessionID, and terminal services SessionState—is also obtained.
- Using the DomainName, UserName, and SessionID of a specific session on the novice computer, the expert queries the novice for a **Remote Assistance Connection** String. The novice starts **Remote Assistance** and returns a Remote Assistance Connection String. The novice is now waiting for the expert to make a peer-to-peer connection to its terminal services session.
- After a Remote Assistance Connection String is obtained by the expert, it is used to make a peer-to-peer Remote Assistance connection to the novice's Terminal Service session on the novice computer.

1.7 Versioning and Capability Negotiation

This document covers versioning issues in the following areas.

Supported Transports: This protocol uses the **DCOM** technology specified in [\[MS-DCOM\]](#), which provides capabilities to query for interface versions.

Protocol Versions: This protocol is composed of the following two primary DCOM interfaces, which are version 0.0: [<2>](#)

- IPCHService
- IRASrv

Both these interfaces offer similar functionality through their methods. An **RPC** client determines if a method is supported by attempting to invoke the method; if the method is not supported, the RPC server returns the "RPC_S_PROCNUM_OUT_OF_RANGE" error, as specified in [\[MS-RPCE\]](#) section 3.1.1.5.5.

When an **expert** attempts to connect using this protocol, it uses only one of these two interfaces.

The **novice** computer implements at least one of these two interfaces, and can implement both interfaces.

The expert computer negotiates for a given set of server functionality by specifying the **UUID** corresponding to the wanted RPC interface when binding to the novice.

1.8 Vendor-Extensible Fields

This protocol does not define any vendor-extensible fields.

1.9 Standards Assignments

No standards assignments have been received for this protocol. All values used in these extensions are in private ranges specified in section [2.1](#).

Parameter	Value	Reference
RPC Interface UUID for IPCHService	833E4200-AFF7-4AC3-AAC2-9F24C1457BCE	IPCHService

Parameter	Value	Reference
RPC Interface UUID for IPCHCollection	833E4100-AFF7-4AC3-AAC2-9F24C1457BCE	IPCHCollection
RPC Interface UUID for ISAFSession	833E41AA-AFF7-4AC3-AAC2-9F24C1457BCE	ISAFSession
RPC Interface UUID for IRASrv	F120A684-B926-447F-9DF4-C966CB785648	IRASrv
CLSID for PCHService object	833E4010-AFF7-4AC3-AAC2-9F24C1457BCE	PCHService
CLSID for RASrv object	3C3A70A7-A468-49B9-8ADA-28E11FCCAD5D	RASrv

2 Messages

2.1 Transport

DCOM [\[MS-DCOM\]](#) is used as the transport protocol. The Remote Assistance Initiation Protocol documented here relies upon DCOM [\[MS-DCOM\]](#) authentication and encryption for all protocol messages.

This protocol MUST use these **universally unique identifier (UUID)** interfaces, as explained in section [1.9](#):

- (833E4200-AFF7-4AC3-AAC2-9F24C1457BCE)
- (833E4100-AFF7-4AC3-AAC2-9F24C1457BCE)
- (833E41AA-AFF7-4AC3-AAC2-9F24C1457BCE)
- (F120A684-B926-447F-9DF4-C966CB785648)

The following two are the **CLSIDs** of the objects implementing these interfaces, as explained in section 1.9:

- (833E4010-AFF7-4AC3-AAC2-9F24C1457BCE)
- (3C3A70A7-A468-49B9-8ADA-28E11FCCAD5D)

2.2 Common Data Types

In addition to **RPC** base types and definitions specified in [\[C706\]](#) and [\[MS-DTYP\]](#), additional data types are defined in the following sections.

There are two version-specific forms of the **Remote Assistance Connection** String, appended with 1 and 2 to differentiate them.

2.2.1 Remote Assistance Connection String 1

The Remote Assistance Connection String 1 contains the **Remote Desktop Protocol** parameters to establish a **Remote Assistance connection**. The Remote Assistance Connection String 1 is a comma-delimited **Unicode string** comprising the following pieces of data. [<3>](#)

```
<ProtocolVersion>,<protocolType>,<machineAddressList>,<assistantAccountPwd>,  
<RASessionID>,<RASessionName>,<RASessionPwd>,<protocolSpecificParms>
```

The elements are defined as follows.

ProtocolVersion: Identifies the protocol version. The value MUST be 65,538.

protocolType: Identifies the protocol type. The value MUST be 1.

machineAddressList: Identifies network address of server machine. It is a semicolon-delimited list of IPAddress(or **computer name**):PortID string pairs. The IPAddress is of type IPV4.

assistantAccountPwd: Password for **Remote Assistance** Account Name. This MUST be set to "".

RASessionID: Remote Assistance session unique identifier.

RASessionName: Remote Assistance session name. This MUST be set to "".

RASessionPwd: Remote Assistance session password. This MUST be set to "*".

protocolSpecificParms: Parameter specific to a Remote Desktop Protocol.

The following is an example of Remote Assistance Connection String 1.

```
65538,1,172.31.243.138:3389;MIKE_HOME:3389,*,Uj7Rp0lU80SibpRw
RZ9+z1vbh7nIgVn89XlAiKp15Vc=,*,*,RcfwecK8dpcT1fjZ6iQ5M0+q7iU=
```

The values for the Remote Assistance Connection String 1 entities in the preceding example are as follows.

```
ProtocolVersion = 65538
protocolType = 1
machineAddressList = 172.31.243.138:3389;MIKE HOME:3389
assistantAccountPwd = *
RASessionID = Uj7Rp0lU80SibpRwRZ9+z1vbh7nIgVn89XlAiKp15Vc=
RASessionName = *
RASessionPwd = *
protocolSpecificParms = RcfwecK8dpcT1fjZ6iQ5M0+q7iU=
```

2.2.2 Remote Assistance Connection String 2

The second type of the **Remote Assistance Connection** String packet is a **Unicode string** in XML format.[<4>](#) The details of this are as follows.

```
<E>
  <A KH="Protocol-specific Parameter" KH2="Protocol-specific Parameter" ID="Authorization
String Identifier" />
  <C>
    <T ID="Transport ID" SID="Session ID">
      <L P="Port" N="Server Name" />
    </T>
  </C>
</E>
```

The novice (server) generates the KH attribute of the Auth String Node <A> in the Remote Assistance Connection String 2. The expert (client) validates the value of KH during the RDP connection sequence.

The KH value is a base64-encoded string of the SHA1 hash of the **PublicKeyBlob** field of the server certificate received in TS_UD_SC_SEC1. The length, in bytes, of the **PublicKeyBlob** is given by the **wPublicKeyBlobLen** field as specified in [\[MS-RDPBCGR\]](#) sections 2.2.1.4.3 and 2.2.1.4.3.1.1.

In addition to the KH attribute, the novice (server) SHOULD [<5>](#) generate the KH2 attribute of the Auth String Node <A> in the Remote Assistance Connection String 2. If this parameter is present in the connection string, the expert (client) validates the value of KH2 during the RDP connection. If this parameter is absent in the connection string, the client uses the KH parameter for validating the server as described in the previous paragraph. The KH2 value is a composite of the hashing algorithm used and the base64 encoded string of the PublicKeyBlob field of the server certificate received in TS_UD_SC_SEC1. The hashing algorithm can be sha256, sha384, or sha512. The hashing algorithm and base64 encoded string of the hash are separated by a colon (:).

The Remote Assistance Connection String 2 starts with the root Node <E>. This root Node contains the following child nodes:

1. The Auth String Node <A> has the following attributes.

Value	Meaning
KH	Parameters specific to a Remote Desktop Protocol
ID	Authstring identifier
KH2	Parameters specific to a Remote Desktop Protocol

2. The Connector Node <C> has child nodes that give information on the underlying Transport used. This Transport Node <T> has the following attributes.

Value	Meaning
ID	Transport Identifier
SID	Session Identifier

3. The Transport Node has Listener child Nodes that give information about the Server IP and port. This Listener node <L> has the following attributes.

Value	Meaning
P	Port: The dynamic port on which the Remote Assistance connection could happen.
N	Server Name: The name/IP address of the server, that is, the novice computer.
U	URI: The full URI if websocket listener is enabled. The U (URI) is used instead of the P (port) attribute. N (server name) attribute is still included.

The following is an example of Remote Assistance Connection String 2:

```
<E>
  <A KH="YiKwWUY8Ioq5NB3wAQHSbs5kwrM="
KH2="sha256:wKSAkAV3sBfa9WpuRFJcP9q1twJc6wOBuoJ9tsyXwpk="
ID="8rYm30RBW8/4dAWoUsWbFCF5jno/7jr5t
NpHQc2goLbw4uuBBJvLsU02YYL1BMg5"/>
  <C>
    <T ID="1" SID="1440550163">
      <L P="49749" N="2001:4898:1a:5:79e2:3356:9b22:3470"/>
      <L P="49751" N="172.31.250.64"/>
    </T>
  </C>
</E>
```

In the preceding example:

1. The Auth String Node <A> has the attribute KH="YiKwWUY8Ioq5NB3wAQHSbs5kwrM=", KH2="sha256:wKSAkAV3sBfa9WpuRFJcP9q1twJc6wOBuoJ9tsyXwpk=", and attribute ID = "8rYm30RBW8/4dAWoUsWbFCF5jno/7jr5tNpHQc2goLbw4uuBBJvLsU02YYLIBMg5". In this example, the KH2 value contains the encoded string "wKSAkAV3sBfa9WpuRFJcP9q1twJc6wOBuoJ9tsyXwpk=" and the hashing algorithm used is SHA256.
2. The Connector Node <C> has one Transport child Node <T> with the following attributes:

ID = "1"

Session ID - SID = "1440550163"

3. The Transport Node has two Listener child Nodes <L> with the following server and port information attributes:

Port - P = "49749"

Server Name - N = "2001:4898:1a:5:79e2:3356:9b22:3470", and

Port - P = "49751"

Server Name - N = "172.31.250.64".

2.2.3 SessionStateEnum

The SessionStateEnum enumeration defines the states of a **terminal services** session.

```
typedef enum
{
    pchActive = 0,
    pchConnected = 1,
    pchConnectQuery = 2,
    pchShadow = 3,
    pchDisconnected = 4,
    pchIdle = 5,
    pchListen = 6,
    pchReset = 7,
    pchDown = 8,
    pchInit = 9,
    pchStateInvalid = 10
} SessionStateEnum;
```

pchActive: The user is logged on and active.

pchConnected: The server is connected to the client.

pchConnectQuery: The server is in the process of connecting to the client.

pchShadow: The session is shadowing another session.

pchDisconnected: The client has disconnected from the session.

pchIdle: The session is waiting for a client to connect.

pchListen: The session is listening for a request for a new connection. No user is logged on to a listener session. A listener session cannot be reset, shadowed, or changed to a regular client session.

pchReset: The session is being reset.

pchDown: The session is down due to an error.

pchInit: The session is initializing.

pchStateInvalid: The session is in an unknown state.

3 Protocol Details

3.1 IPCHService Remote Assistance Server Details

The **Remote Assistance** server provides methods that allow a client to:

- Get the collection of the **terminal services** sessions on the remote **novice** computer.
- Get a **Remote Assistance Connection** String for a specific Terminal Server session.

The following sections specify server details of the IPCHService interface of the Remote Assistance Initiation Protocol, including abstract data models, timers, and message processing rules.

3.1.1 Abstract Data Model

No abstract data model is used.

3.1.2 Timers

No timers or time-out periods are associated with this section of the protocol.

3.1.3 Initialization

The server **MUST** listen on the **well-known endpoint** defined for this **RPC** interface. For more information, see section [2.1](#).

3.1.4 Message Processing Events and Sequencing Rules

3.1.4.1 IPCHService

The IPCHService interface is implemented by the **novice** to allow the **expert** to request a **Remote Assistance Connection** String.[<6>](#)

The **UUID** for this interface is: "833E4200-AFF7-4AC3-AAC2-9F24C1457BCE".

Methods in RPC opnum order:

- **Opnums** 0, 1, and 2 are reserved for the IUnknown_QueryInterface, AddRef, and Release methods used by the standard COM IUnknown interface specified in [\[MS-DCOM\]](#) section 3.1.1.5.8.
- Opnums 3 and 4 are not used across the network. These opnums are reserved and **MUST NOT** be used.[<7>](#)
- Opnums 5 and 6 are reserved for the GetIDsOfNames and Invoke methods in the IDispatch interface specified in [\[MS-OAUT\]](#) section 3.1.
- Opnums 7 through 18 and opnum 21 are not used by this protocol.

Method	Description
RemoteConnectionParms	Gets the Remote Assistance connection parameters for a specified UserName, DomainName, and SessionID triple. Opnum: 19
RemoteUserSessionInfo	Returns the collection of the terminal services sessions on the remote novice computer. All the terminal services session information is returned as a standard

Method	Description
	IPCHCollection interface. The members of this collection are objects of type ISAFSession. ISAFSession includes the DomainName, UserName, SessionID, and User SessionState for each session. Opnum: 20

3.1.4.1.1 RemoteConnectionParms (Opnum 19)

The RemoteConnectionParms method gets the **Remote Assistance connection** parameters for a specific UserName, DomainName, and SessionID triple.

```
[id(DISPID_PCH_SVC_REMOTECONNECTIONPARMS)] HRESULT RemoteConnectionParms(
    [in] BSTR bstrUserName,
    [in] BSTR bstrDomainName,
    [in] long lSessionID,
    [in] BSTR bstrUserHelpBlob,
    [out, retval] BSTR* pbstrConnectionString
);
```

bstrUserName: The UserName part of the DomainName\UserName string corresponding to the **terminal services** session for which the client is requesting a Remote Assistance Connection String.

bstrDomainName: The DomainName part of the DomainName\UserName string corresponding to the terminal services session for which the client is requesting a Remote Assistance Connection String.

lSessionID: Identifier of the terminal services session for which the client is requesting a Remote Assistance Connection String.

bstrUserHelpBlob: A semicolon-delimited string that contains the **domain** and user names of the **expert** requesting a Remote Assistance Connection String. The format of the string is as follows.

```
<Length of the string>;
<"UNSOLICITED=1"> (Note: no semicolon)
<Length of the DomainName\UserName string>;
<ID=DomainName\UserName> (Note: characters 'ID=' count toward string length)
```

The following is an example.

```
"13;UNSOLICITED=118;ID=EXDOMAIN\EXUSER"
```

pbstrConnectionString: A pointer to a Remote Assistance Connection String for the requested session.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure. If the UserName and DomainName are valid BSTRs, the return code is one listed in the following table. If the UserName and DomainName are invalid BSTRs, the HRESULT value returned is the corresponding HRESULT to the system error code ERROR_NONE_MAPPED.

Return value/code	Description
0x00000000	The call was successful.

Return value/code	Description
S_OK	
0x80070005 E_ACCESSDENIED	General access denied error. <8>
0x8007000E E_OUTOFMEMORY	Out of memory.
0x800704EC ERROR_ACCESS_DISABLED_BY_POLICY	The program cannot be opened because of a software restriction policy. For more information, contact the system administrator.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2 RemoteUserSessionInfo (Opnum 20)

The RemoteUserSessionInfo method returns the collection of the **terminal services** sessions on the remote **novice** machine. All the terminal services session information is returned as a standard IPCHCollection interface. The members of this collection are objects of type ISAFSession. ISAFSession includes the DomainName, SessionID, SessionState, and UserName for each session.

```
[id(DISPID_PCH_SVC__REMOTEUSERSESSIONINFO)] HRESULT RemoteUserSessionInfo(
    [out, retval] IPCHCollection** pVal
);
```

pVal: A pointer to an [IPCHCollection](#) interface containing terminal services sessions information on the server.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80070005 E_ACCESSDENIED	General access denied error. <9>
0x8007000E E_OUTOFMEMORY	Out of memory.
0x800704EC ERROR_ACCESS_DISABLED_BY_POLICY	The program cannot be opened because of a software restriction policy. For more information, contact the system administrator.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.1 IPCHCollection

The IPCHCollection interface provides methods to enumerate the elements of a collection.

The **UUID** for this interface is: "833E4100-AFF7-4AC3-AAC2-9F24C1457BCE".

Opnums 3 and 4 are not used across the network. These opnums are reserved and MUST NOT be reused by non-Microsoft implementations. <10>

Methods in RPC Opnum Order

Method	Description
_NewEnum	Creates a copy of the collection. Opnum: 7
Item	Retrieves an element. Opnum: 8
Count	Retrieves the number of elements in the collection. Opnum: 9

Opnums 0, 1, and 2 are reserved for the IUnknown_QueryInterface, AddRef, and Release methods used by the standard COM IUnknown interface, as specified in [\[MS-DCOM\]](#) section 3.1.1.5.8. Opnums 5 and 6 are reserved for the GetIDsOfNames and Invoke methods in the IDispatch interface, as specified in [\[MS-OAUT\]](#) section 3.1.

3.1.4.1.2.1.1 _NewEnum (Opnum 7)

The _NewEnum method creates a copy of the collection.

```
[propget, id(DISPID_NEWENUM)] HRESULT _NewEnum(  
    [out, retval] IUnknown** pVal  
);
```

pVal: A pointer to a pointer to the **IUnknown** interface of a new copy of the collection.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80004003 E_POINTER	The method failed due to an invalid pointer.
0x8007000E E_OUTOFMEMORY	The method was unable to allocate required memory.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.1.2 Item (Opnum 8)

The Item method retrieves an element.

```
[propget, id(DISPID_VALUE)] HRESULT Item(  

```

```

[in] long vIndex,
[out, retval] VARIANT* ppEntry
);

```

vIndex: One-indexed number of the element to retrieve.

ppEntry: A pointer to the element at *vIndex*, of type VARIANT as defined in [\[MS-OAUT\]](#) section 2.2.29.2.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80004003 E_POINTER	The method failed due to an invalid pointer.
0x8007000E E_OUTOFMEMORY	The method was unable to allocate required memory.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.1.3 Count (Opnum 9)

The Count method retrieves the number of elements in the collection.

```

[propget, id(DISPID_PCH_COL_COUNT)] HRESULT Count(
[out, retval] long* pVal
);

```

pVal: A pointer to the number of elements in the collection.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80004003 E_POINTER	The method failed due to an invalid pointer.
0x8007000E E_OUTOFMEMORY	The method was unable to allocate required memory.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2 ISAFSession

The ISAFSession interface is implemented by servers to describe sessions.

The **UUID** for this interface is:

"833E41AA-AFF7-4AC3-AAC2-9F24C1457BCE".

Opnums 3 and 4 are not used across the network. These opnums are reserved and MUST NOT be reused by non-Microsoft implementations. <11>

Methods in RPC Opnum Order

Method	Description
SessionID (Get)	Retrieves the identifier number of the session. Opnum: 7
SessionID (Set)	Sets the identifier number of the session. Opnum: 8
SessionState (Get)	Retrieves the state of the session. Opnum: 9
SessionState (Set)	Sets the state of the session. Opnum: 10
DomainName (Get)	Retrieves the domain name for the session user. Opnum: 11
DomainName (Set)	Sets the domain name for the session user. Opnum: 12
UserName (Get)	Retrieves the user name for the session. Opnum: 13
UserName (Set)	Sets the user name for the session. Opnum: 14

Opnums 0, 1, and 2 are reserved for the IUnknown_QueryInterface, AddRef, and Release methods used by the standard COM IUnknown interface, as specified in [\[MS-DCOM\]](#) section 3.1.1.5.8. Opnums 5 and 6 are reserved for the GetIDsOfNames and Invoke methods in the IDispatch interface, as specified in [\[MS-OAUT\]](#) section 3.1.

3.1.4.1.2.2.1 DomainName (Get) (Opnum 11)

The DomainName (Get) method retrieves the **domain name** for the session user.

```
[propget, id(DISPID_SAF_SESS__DOMAINNAME)] HRESULT DomainName(  
    [out, retval] BSTR* pVal  
);
```

pVal: Pointer to the domain name for the session user.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80004003 E_POINTER	The method failed due to an invalid pointer.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.2 DomainName (Set) (Opnum 12)

The DomainName (Set) method sets the **domain name** for the session user.

```
[propput, id(DISPID_SAF_SESS__DOMAINNAME)] HRESULT DomainName(
    [in] BSTR pVal
);
```

pVal: Domain name to assign for the session user.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.3 SessionID (Get) (Opnum 7)

The SessionID (Get) method retrieves the identifier number of the session.

```
[propget, id(DISPID_SAF_SESS__SESSIONID)] HRESULT SessionID(
    [out, retval] DWORD* pVal
);
```

pVal: Pointer to the identifier number of the session.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80000003 E_INVALIDARG	One or more arguments are invalid.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.4 SessionID (Set) (Opnum 8)

The SessionID (Set) method sets the identifier number of the session.

```
[propput, id(DISPID_SAF_SESS__SESSIONID)] HRESULT SessionID(  
    [in] DWORD pVal  
);
```

pVal: Identifier number of the session to assign.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.5 SessionState (Get) (Opnum 9)

The SessionState (Get) method retrieves the state of the session.

```
[propget, id(DISPID_SAF_SESS__SESSIONSTATE)] HRESULT SessionState(  
    [out, retval] SessionStateEnum* pVal  
);
```

pVal: A pointer to an integer that represents the state of the session. The integer MUST be one of the values of [SessionStateEnum](#), as specified in section 2.2.3.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80000003 E_INVALIDARG	One or more arguments are invalid.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.6 SessionState (Set) (Opnum 10)

The SessionState (Set) method sets the state of the session.

```
[propput, id(DISPID_SAF_SESS__SESSIONSTATE)] HRESULT SessionState(  
    [in] SessionStateEnum pVal  
);
```

pVal: An integer that represents the state of the session. The integer MUST be one of the values of the [SessionStateEnum](#) enumeration, as specified in section 2.2.3.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80000003 E_INVALIDARG	One or more arguments are invalid.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.7 UserName (Get) (Opnum 13)

The UserName (Get) method retrieves the user name for the session.

```
[propget, id(DISPID_SAF_SESS__USERNAME)] HRESULT UserName(  
    [out, retval] BSTR* pVal  
);
```

pVal: Pointer to the user name for the session.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80004003 E_POINTER	The method failed due to an invalid pointer.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.4.1.2.2.8 UserName (Set) (Opnum 14)

The UserName (Set) method sets the user name for the session.

```
[propput, id(DISPID_SAF_SESS__USERNAME)] HRESULT UserName(  

```

```
[in] BSTR pVal  
);
```

pVal: User name to assign for the session.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.1.5 Timer Events

No timer events are required beyond the events maintained in the underlying **RPC** transport (see section [2.1](#)).

3.1.6 Other Local Events

No additional local events are used beyond the events maintained in the underlying **RPC** transport (see section [2.1](#)).

3.2 IPCHService Remote Assistance Client Details

The following sections specify client details of the Remote Assistance Initiation Protocol, including abstract data models, timers, and message processing rules.

3.2.1 Abstract Data Model

The **expert** MUST specify the IP address or **FQDN** of the **novice** computer to use this protocol.

Using the specified IP address or FQDN of the novice computer, the expert connects to the novice computer and invokes the [RemoteUserSessionInfo](#) method of IPCHService on the remote novice computer and gets back a collection containing the DomainName, UserName, SessionID and SessionState of the **terminal services** sessions from the novice computer.

The expert selects the terminal services session where **Remote Assistance** is to be offered and by specifying the terminal services Session's DomainName, UserName and SessionID, invokes the [RemoteConnectionParams](#) method of IPCHService on the remote novice computer to obtain the **Remote Assistance Connection** String.

3.2.2 Timers

No protocol timers are required other than those internal ones used in **RPC** to implement resiliency to network outages, as specified in [\[MS-RPCE\]](#).

3.2.3 Initialization

The client creates an **RPC** association (or binding) to the server RPC before an RPC method is called. The client can either create a separate association for each method invocation, or it can reuse an association for multiple invocations.

3.2.4 Message Processing Events and Sequencing Rules

The following list shows the sequence of steps involved in a **Remote Assistance Connection** String request:

1. The client initiates the conversation with the server by performing DCOM activation, as specified in [\[MS-DCOM\]](#) section 3.2.4.1.1, of the **PCHService CLSID** specified in section [1.9](#). As a result of the activation, the client gets the **PCHService**'s interface from the server.
2. The client then obtains an additional **IPCHService** interface from the **PCHService** interface as specified in [\[MS-DCOM\]](#) section 3.2.4.4.3.
3. The client then invokes the [RemoteUserSessionInfo](#) method of the **IPCHService** interface to retrieve an **ISAFSession** interface pointers' collection of type **IPCHCollection**.
4. Enumerate the **ISAFSession** interfaces of the server by repeatedly calling the Item method of the **IPCHCollection** interface. The collected list of session information is presented to the **Remote Assistance** user, by means of an implementation-specific user interface, to decide which session to connect to.
5. From the collected **ISAFSession** interface values, a session is chosen by the Remote Assistance user. The SessionID and corresponding DomainName and UserName are passed in a call to the [RemoteConnectionParms](#) method of the **IPCHService** interface. This call returns the Remote Assistance Connection String for the specified session. Sessions with no DomainName or UserName MUST be ignored.
6. This Remote Assistance Connection String is valid to begin the Remote Assistance connection.

3.2.5 Timer Events

No protocol timer events are required on the client other than the events maintained in the underlying **RPC** transport.

3.2.6 Other Local Events

No additional local events are used on the client other than the events maintained in the underlying **RPC** transport.

3.3 IRASrv Remote Assistance Server Details

The **Remote Assistance** Server provides methods to allow a client to:

- Get the collection of the **terminal services** sessions on the remote **novice** computer.
- Get a **Remote Assistance Connection** String for a specified user session on the remote novice computer.

The following sections specify server details of the Remote Assistance Initiation Protocol, including abstract data models, timers, and initialization.

3.3.1 Abstract Data Model

No abstract data model is used.

3.3.2 Timers

No protocol timers are required other than those internal ones used in **RPC** to implement resiliency to network outages, as specified in [\[MS-RPCE\]](#).

3.3.3 Initialization

The server **MUST** listen on the **well-known endpoint** defined for this **RPC** interface. For more information, see section [2.1](#).

3.3.4 Message Processing Events and Sequencing Rules

3.3.4.1 IRASrv

The IRASrv interface is implemented by the **novice** computer to allow the **expert** computer to request a **Remote Assistance Connection String**.[<12>](#)

The **UUID** for this interface is: "F120A684-B926-447F-9DF4-C966CB785648".

Opnums 3 and 4 are not used across the network. These opnums are reserved and **MUST NOT** be reused by non-Microsoft implementations.[<13>](#)

Methods in RPC Opnum Order

Method	Description
GetNoviceUserInfo	Retrieves the Remote Assistance Connection String information for a specified terminal services session. Opnum: 7
GetSessionInfo	Returns the collection of the terminal services sessions on the remote novice computer. This method also returns the number of terminal services sessions running on the novice computer. All the terminal services session information is returned as a SAFEARRAY. The elements of this SAFEARRAY are strings with information about the DomainName, UserName, and SessionID for each terminal services session. Opnum: 8

Opnums 0, 1, and 2 are reserved for the IUnknown_QueryInterface, AddRef, and Release methods used by the standard COM IUnknown interface, as specified in [\[MS-DCOM\]](#) section 3.1.1.5.8. Opnums 5 and 6 are reserved for the GetIDsOfNames, and Invoke methods in the IDispatch interface, as specified in [\[MS-OAUT\]](#) section 3.1.

3.3.4.1.1 GetNoviceUserInfo (Opnum 7)

The GetNoviceUserInfo method is received by the server in an RPC_REQUEST packet. The method is received in the **terminal services** session as specified by the Client. In response, the server returns the Remote Assistance Connection String 2 for the specified terminal services session.

```
[id(1), helpstring("method GetNoviceUserInfo")] HRESULT GetNoviceUserInfo(  
    [in, out] LPWSTR* szName  
);
```

szName: A pointer to a NULL-terminated **Unicode string** that contains the Remote Assistance Connection String 2 for the specified terminal services session.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.
0x80004003 E_POINTER	The method failed due to an invalid pointer for szName.
0x8007000E E_OUTOFMEMORY	Out of memory.
0x80070052 ERROR_CANNOT_MAKE	An instance of Remote Assistance is already running on the novice's machine.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.3.4.1.2 GetSessionInfo (Opnum 8)

The GetSessionInfo method is received by the server in an RPC_REQUEST packet. In response, the server returns the **terminal services** session information for the various sessions on the computer. The terminal services session information is returned as a SAFEARRAY of BSTRs. Each BSTR contains the DomainName, UserName and SessionID in the format DomainName\UserName:SessionID.

This method also returns the count of the total number of sessions.

This method does not return Idle and Disconnected terminal services sessions. Any null values returned in the SAFEARRAY can be ignored.

```
[id(2), helpstring("method GetSessionInfo")] HRESULT GetSessionInfo(  
    [in, out] SAFEARRAY(BSTR)* UserNames,  
    [in, out] INT* Count  
);
```

UserNames: A pointer to a SAFEARRAY, as specified in [\[MS-OAUT\]](#), of BSTRs containing the terminal services session information. Each BSTR element in the array contains the DomainName, UserName, and SessionID in the format DomainName\UserName:SessionID. This is returned to the **expert**.

Count: A pointer to an INT that returns the number of terminal services sessions on the **novice**.

Return Values: A signed 32-bit value indicating return status. This method MUST return zero to indicate success, or an HRESULT error value (as specified in [\[MS-ERREF\]](#) section 2.1.1) to indicate failure.

Return value/code	Description
0x00000000 S_OK	The call was successful.

Return value/code	Description
0x80004003 E_POINTER	The method failed due to an invalid pointer.
0x8007000E E_OUTOFMEMORY	The method was unable to allocate required memory.

Exceptions Thrown:

No exceptions are thrown beyond those thrown by the underlying **RPC** protocol [\[MS-RPCE\]](#).

3.3.5 Timer Events

No protocol timer events are required on the client other than the events maintained in the underlying **RPC** transport.

3.3.6 Other Local Events

No additional local events are used other than the events maintained in the underlying **RPC** transport.

3.4 IRASrv Remote Assistance Client Details

The following sections specify client details of the Remote Assistance Initiation Protocol, including abstract data models, timers, and message processing rules.

3.4.1 Abstract Data Model

The **expert** computer MUST specify the IP address or **FQDN** of the **novice** computer to use this protocol.

Using the specified IP Address or FQDN of the novice computer, the expert computer connects to the novice computer and invokes the [GetSessionInfo](#) method of IRASrv on the remote novice computer. GetSessionInfo returns a SAFEARRAY of BSTRs containing the DomainName, UserName and SessionID of the **terminal services** sessions from the novice computer.

The expert computer selects the terminal services session where **Remote Assistance** is to be offered and invokes the [GetNoviceUserInfo](#) method of the IRASrv interface running on the remote novice computer. The GetNoviceUserInfo method returns the **Remote Assistance Connection** String for the terminal services session where it is being invoked.

3.4.2 Timers

No protocol timers are required other than those internal ones used in **RPC** to implement resiliency to network outages, as specified in [\[MS-RPCE\]](#).

3.4.3 Initialization

The client creates an **RPC** association (or binding) to the server RPC before an RPC method is called. The client can either create a separate association for each method invocation, or it can reuse an association for multiple invocations.

3.4.4 Message Processing Events and Sequencing Rules

The following list shows the sequence of steps involved in a **Remote Assistance Connection** String request:

1. The client initiates the conversation with the server by performing DCOM activation, as specified in [\[MS-DCOM\]](#) section 3.2.4.1.1, of the **RASrv CLSID** specified in section [1.9](#). As a result of the activation, the client gets the **RASrv**'s interface from the server.
2. The client then obtains an additional **IRASrv** interface from the **RASrv** interface as specified in [\[MS-DCOM\]](#) section 3.2.4.4.3.
3. The client then invokes the [GetSessionInfo](#) method of the **IRASrv** interface to retrieve the **terminal services** session information as a SAFEARRAY of BSTRs and the count of the number of sessions.
4. From the collected terminal services session entries, choose a session and use the corresponding SessionID to invoke the [GetNoviceUserInfo](#) method of **IRASrv** in the specified session. This call returns the Remote Assistance Connection String for the specified session.
5. This Remote Assistance Connection String is valid to begin the Remote Assistance connection from the **expert** computer to the **novice** computer.

3.4.5 Timer Events

No protocol timer events are required on the client other than the events maintained in the underlying **RPC** transport.

3.4.6 Other Local Events

No additional local events are used on the client other than the events maintained in the underlying **RPC** transport.

4 Protocol Examples

The following example demonstrates the client and server for [IPCHService](#) interface, using the [RemoteUserSessionInfo](#) and [RemoteConnectionParms](#) methods.

1. The client (**expert**) calls the RemoteUserSessionInfo method on the server.
2. The server (**novice**) returns with code 0x00000000 and with the [IPCHCollection](#) interface pointer.
3. The client gets the count of the **terminal services** sessions in the collection by calling the [Count](#) method on the IPCHCollection interface.
4. The client indexes the collection from 1 through count using the [Item](#) method, and for each item in the collection retrieves the [ISAFSession](#) interface pointer.
5. From the ISAFSession interface the client gets:
 1. The DomainName using the [DomainName\(Get\)](#) method.
 2. The UserName using the [UserName\(Get\)](#) method.
 3. The SessionID using the [SessionID\(Get\)](#) method.
6. The client selects a DomainName, UserName, and SessionID.
7. The client formats the UserBlob, for example: "13;UNSOLICITED=119;ID=TESTDOMAIN\Admin".
8. The client calls the RemoteConnectionParms method using the UserName, DomainName, SessionID, and UserBlob.
9. The server returns with code 0x00000000, and with the **Remote Assistance Connection** String.

5 Security

5.1 Security Considerations for Implementers

This protocol relies on the security features provided by **DCOM** [\[MS-DCOM\]](#). Review the security considerations listed in [\[MS-RPCE\]](#) section 5.1, as these are also valid for DCOM and DCOM-based protocols. [<14>](#)

5.2 Index of Security Parameters

There are no security parameters for this protocol.

6 Appendix A: Remote Assistance Invitation File Format

The purpose of the Remote Assistance Initiation Protocol is to obtain the **Remote Assistance Connection** String using **DCOM**. However, the implementer is free to use alternative approaches to obtain the connection string. One such alternative approach implemented in Windows is to transmit the Remote Assistance Invitation File over email as an XML file. [<15>](#15)

The **expert** computer parses the Remote Assistance Invitation File to extract the Remote Assistance Connection String and other information.

There are two version-specific types of the Remote Assistance Invitation File.

The following is a sample Remote Assistance Invitation File of the first type: [<16>](#16)

```
<?xml version="1.0" encoding="Unicode" ?>
  <UPLOADINFO TYPE="Escalated">
    <UPLOADDATA
      USERNAME="jeff"
      RCTICKET="65538,1,192.168.1.65:3389;jeff_xp:3389,*,ot9B5Ut8n6FmiIOr2Aa91
      5WwuLcMdtNl5AoXFiA4wLg=,*,*,5nKH3X0Ikre0jjL9SaRlfN10p9o="
      RCTICKETENCRYPTED="1"
      DtStart="1160080069"
      DtLength="60"
      PassStub="o2*5GdBARK_JBB"
      L="0" />
    </UPLOADDATA>
  </UPLOADINFO>
```

The file contains the following pieces of information:

Value	Meaning
UPLOADINFO	The UPLOADINFO TYPE is set to "Escalated".
UPLOADDATA	The UPLOADDATA contains the following information: USERNAME: The name of the Remote Assistance novice computer. RCTICKET: The Remote Assistance Connection String. RCTICKETENCRYPTED: Indicates whether the ticket identified by RCTICKET is encrypted. RCTICKETENCRYPTED = 1 means the ticket is encrypted, zero means the ticket is not encrypted. DtStart: The time when the Remote Assistance Connection String was created. This time is the number of seconds elapsed since UTC 1/1/70. DtLength: The duration for which the Remote Assistance Connection String is valid. This is expressed in minutes. If the expert computer cannot connect to the novice computer successfully within this time, Remote Assistance closes on the novice computer. PassStub: The encrypted novice computer's password string. When the Remote Assistance Connection String is sent as a file over email, to provide additional security, a password is used. <17> L: Indicates whether the novice computer is connected via a modem. L = 1 means MODEM is used, 0 means high-speed connectivity is used.

The following diagram shows the password encryption flow for specific implementations. [<18>](#18)

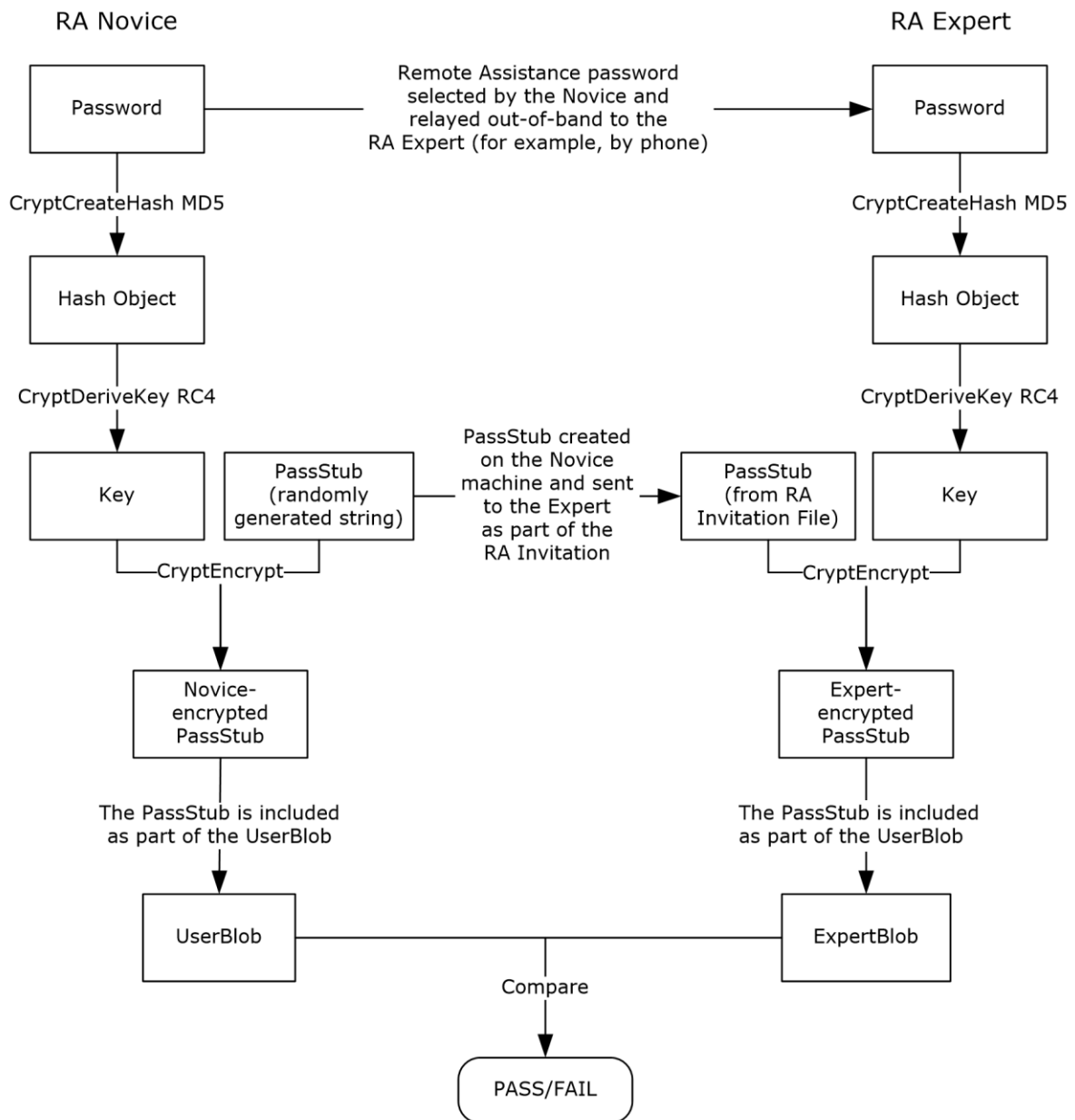


Figure 2: Password encryption flow

The following is a sample Remote Assistance Invitation File of the second type: [<19>](#)

```

<?xml version="1.0" encoding="Unicode" ?>
<UPLOADINFO TYPE="Escalated">
  <UPLOADDATA
    USERNAME="jeff"
    LHTICKET="3CD9C8A0DB0628410E91EC277CAEB705E5422CE1DA55E0C118155A8BA465C3E81552DC
    B85D03F6A7F2F930C44C1D097239DB8B47339A01D4392F4F05985106757148AEA4C6832BA2AC7C46
  >

```

```

0B958BD4F7966DBBC76E72F6F47FEE1AC50844D654D2D86A760854286F9DAA3823F0346D41063C7
6378535688017C2D00D263AC187F6BE26FDB854B01E1BC8E4328F54163DB2E901D3805E0D6CF2593
7A2D43C959F51809124DA2E70807A737323968644CB8BC56ECD43AAA40B3B2BA7021198D98AA4D
5B9818095053C0104A52743343489AD1E12AC0CB7001E56910718B9A8155A60AFF3CC26D2B163629
46C32F7F9C22AE844D731740301FBF5951FCF765E052D793F526603AA6B7F86C0697BD02FEF32A8F
4031E30AD55F9752FCA3DB60F8A12466D256F29B22131C1D8ED43E9ACA2BBD172C27D1FD284F825E
E4E65F2D201E042C1C4DEE6C06522A3F015036F7603AAF16D8A6CB595E22CA80572C91F9E163028C
AA4A3E7EA0152045BDAAE8C1221283FF2E23CC53D34870348D6D9CBD5B93411D4F9BB8180062D6B
573CF2D428FDB1CA7142885F6B2A966A149E19F5D00E22E18A3802612521C126455B675F4D7E12C3
6B861C19A2795CE87ED592CCCB735A081E428CC50BC23B460794B53601221FFCCCB458090DEB9B59
19C942A6FD6937F49F9951A9CD416E8E356A6293443EC11798DE204CC67B652C5B62491E37455098
764EE6727AB827411A712C62012026C71BC408D09F6B5FEEB85BF9DE434A5A2BB6F514C4BD06
4D643F0A0EEAE46CCACA4994A6E7CEC42B70EC020DD2CB42058DC919EBE2CBCBE34575F4A40D47D1
EEA092653FC6723B2EC2E88231E46E98598218AA305810360CA972C935B5BCB73769197FD78B835C
3A63A7D603E6CB51F6E84B377D731FEF38C6A9FE68640B58506486C76C33B7F53176E7A52D753DD2
EAE34FB2812663D94B5F6BEF6C878BCD7D41F1983A5B87F1F797D4D7504F9C83A9FE661EBBD57FB
3A01CF82D2E7FF01AE38F70EB8FA8A77B2A4DAB5BDA0E0B458A8FC51F3A354AB9F104DD7B91144D5
E8589F319CDB5ABD4A2B5DBB4B9C43E74309DE5C30102E7165507C5B2A5E613E30F784A0540E206C
38965F1869EE53BD0CC8056B324C242F4DFD5D70EB55082F5A9F5513164D1097536037AB6964DBF4
D3425CBC5F2564B8DBC13889782BFE2C3DB391992781A80B187F6DAE15D643C85BAB3E12B535FEB0
E0BF79FF58"
RCTICKET="65538,1,172.31.244.101:55646,*,BnrZvG4FglMwHBhZgo7SkJEqD90DrPyPnxtC/lv
UcczDCZJacjm0w80gKyzCHTtc,*,*,VasNb+Ymg1mvJ/AJWSh56qq7pk4="
  RCTICKETENCRYPTED="1"
  DtStart="1160080069"
  DtLength="60"
  PassStub="fg^2IkiL*z3j4U"
  L="0" />
</UPLOADINFO>

```

The file contains the following pieces of information:

Value	Meaning
UPLOADINFO	The UPLOADINFO TYPE MUST be "Escalated".
UPLOADDATA	The UPLOADDATA contains the following information: USERNAME: The name of the Remote Assistance novice. LHTICKET: The encrypted Remote Assistance Connection String 2 where each byte is converted to two HEX numbers<20> RCTICKET: The version-specific<21> Remote Assistance Connection String. RCTICKETENCRYPTED: Indicates whether the ticket identified by RCTICKET is encrypted. RCTICKETENCRYPTED = 1 means the ticket is encrypted, zero means the ticket is not encrypted. DtStart: The time when the Remote Assistance Connection String was created. This time is the number of seconds elapsed since UTC 1/1/70. DtLength: The duration for which the Remote Assistance Connection String is valid. This is expressed in minutes. If the expert computer cannot connect to the novice computer successfully within this time, Remote Assistance closes on the novice computer. PassStub: The encrypted novice computer password string. When the Remote Assistance Connection String is sent as a file over email, to provide additional security, a password is used. L: Indicates whether the novice computer is connected via a modem. L = 1 means MODEM is used, 0 means high-speed connectivity is used.

The following diagram shows the password encryption flow for specific implementations.<22>

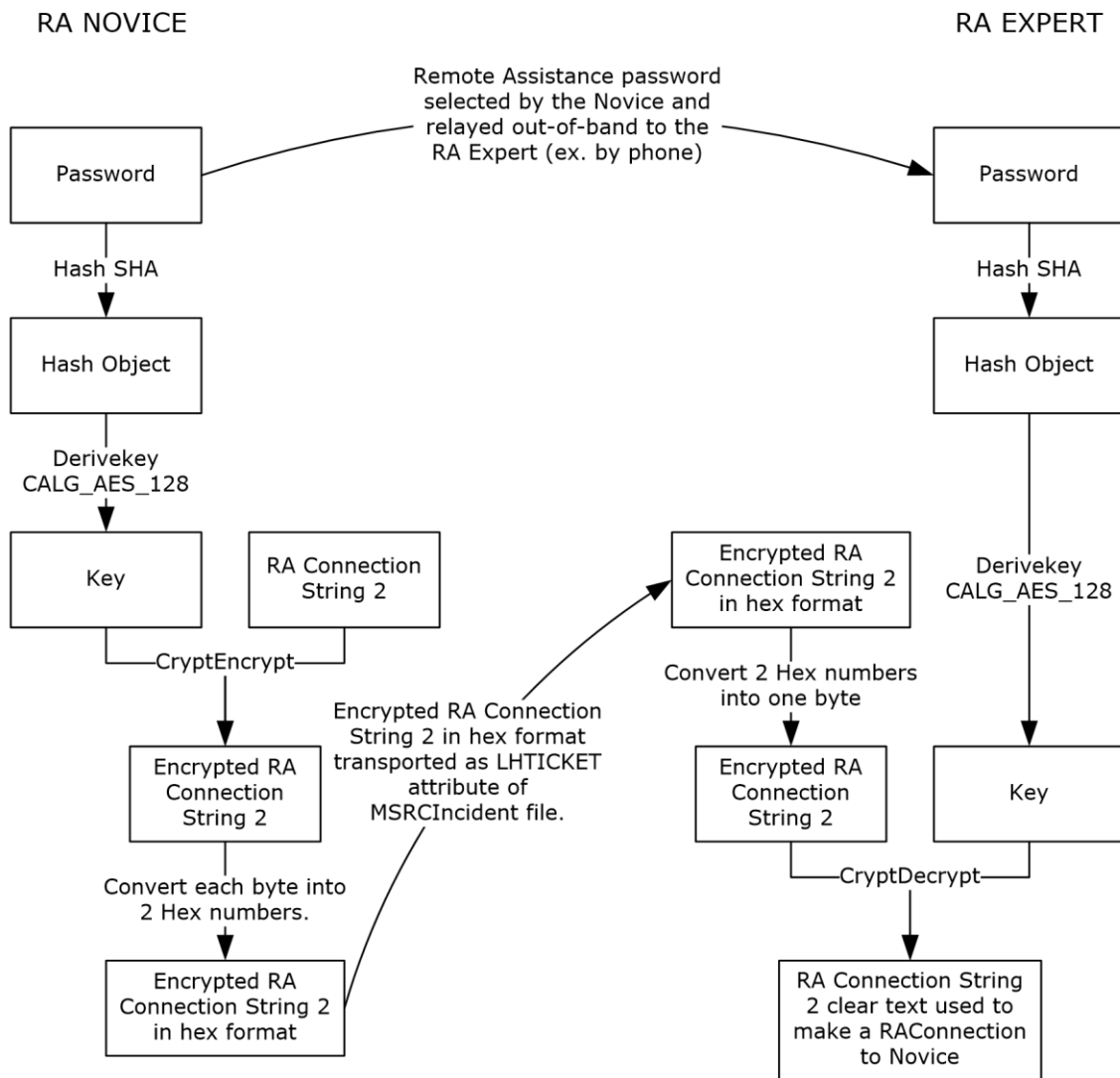


Figure 3: Password encryption flow

7 Appendix B: Full IDL

For ease of implementation, the full IDL is provided below.

This IDL imports the IDL from the OLE Automation Protocol [\[MS-OAUT\]](#), Appendix A, to provide support for the type definitions VARIANT and SAFEARRAY.

```
import "ms-oaut.idl";
#define SAFEARRAY(type) SAFEARRAY
#define DISPID_PCH_BASE 0x08010000
#define DISPID_PCH_BASE_COL (DISPID_PCH_BASE + 0x0000)
#define DISPID_PCH_HELPSVC_BASE 0x08000000
#define DISPID_PCH_HELPSVC_BASE_SVC (DISPID_PCH_HELPSVC_BASE + 0x0000)
#define DISPID_PCH_SVC_REMOTECONNECTIONPARMS (DISPID_PCH_HELPSVC_BASE_SVC + 0x0040)
#define DISPID_PCH_SVC_REMOTEUSERSESSIONINFO (DISPID_PCH_HELPSVC_BASE_SVC + 0x0041)
#define DISPID_PCH_COL_COUNT (DISPID_PCH_BASE_COL + 0x0000)

#define DISPID_SAF_BASE 0x08020000
#define DISPID_SAF_BASE_RCD (DISPID_SAF_BASE + 0x0B00)
#define DISPID_SAF_BASE_USER (DISPID_SAF_BASE + 0x0C00)
#define DISPID_SAF_BASE_SESS (DISPID_SAF_BASE + 0x0D00)
#define DISPID_SAF_USER_DOMAINNAME (DISPID_SAF_BASE_USER + 0x0010)
#define DISPID_SAF_USER_USERNAME (DISPID_SAF_BASE_USER + 0x0011)
#define DISPID_SAF_SESS_SESSIONID (DISPID_SAF_BASE_SESS + 0x0010)
#define DISPID_SAF_SESS_SESSIONSTATE (DISPID_SAF_BASE_SESS + 0x0011)
#define DISPID_SAF_SESS_DOMAINNAME (DISPID_SAF_BASE_SESS + 0x0012)
#define DISPID_SAF_SESS_USERNAME (DISPID_SAF_BASE_SESS + 0x0013)

typedef enum
{
    pchActive = 0,
    pchConnected = 1,
    pchConnectQuery = 2,
    pchShadow = 3,
    pchDisconnected = 4,
    pchIdle = 5,
    pchListen = 6,
    pchReset = 7,
    pchDown = 8,
    pchInit = 9,
    pchStateInvalid = 10
} SessionStateEnum;

[
    object,
    uuid(833E4100-AFF7-4AC3-AAC2-9F24C1457BCE),
    dual,
    oleautomation,
    helpstring("IPCHCollection Interface"),
    pointer_default(unique)
]

interface IPCHCollection : IDispatch
{
    [propget, id(DISPID_NEWENUM)] HRESULT NewEnum(
        [out, retval] IUnknown* *pVal );
    [propget, id(DISPID_VALUE)] HRESULT Item (
        [in] long vIndex,
        [out, retval] VARIANT *ppEntry );
    [propget, id(DISPID_PCH_COL_COUNT)] HRESULT Count (
        [out, retval] long *pVal );
};

[
```

```

        object,
        uuid(833E4200-AFF7-4AC3-AAC2-9F24C1457BCE),
        dual,
        oleautomation,
        helpstring("IPCHService Interface"),
        pointer default(unique)
    ]
interface IPCHService : IDispatch
{
    HRESULT Opnum7NotUsedByProtocol(void);
    HRESULT Opnum8NotUsedByProtocol(void);
    HRESULT Opnum9NotUsedByProtocol(void);
    HRESULT Opnum10NotUsedByProtocol(void);
    HRESULT Opnum11NotUsedByProtocol(void);
    HRESULT Opnum12NotUsedByProtocol(void);
    HRESULT Opnum13NotUsedByProtocol(void);
    HRESULT Opnum14NotUsedByProtocol(void);
    HRESULT Opnum15NotUsedByProtocol(void);
    HRESULT Opnum16NotUsedByProtocol(void);
    HRESULT Opnum17NotUsedByProtocol(void);
    HRESULT Opnum18NotUsedByProtocol(void);
    [id(DISPID_PCH_SVC_REMOTECONNECTIONPARMS)] HRESULT RemoteConnectionParms(
        [in] BSTR bstrUserName,
        [in] BSTR bstrDomainName,
        [in] long lSessionID,
        [in] BSTR bstrUserHelpBlob,
        [out, retval] BSTR *pbstrConnectionString );
    [id(DISPID_PCH_SVC_REMOTEUSERSESSIONINFO)] HRESULT RemoteUserSessionInfo(
        [out, retval] IPCHCollection* *pVal );
    HRESULT Opnum21NotUsedByProtocol(void);
};

[
    object,
    uuid(833E41AA-AFF7-4AC3-AAC2-9F24C1457BCE),
    dual,
    oleautomation,
    helpstring("ISAFSession Interface"),
    pointer_default(unique)
]

interface ISAFSession : IDispatch
{
    [propget, id(DISPID_SAF_SESS__SESSIONID )] HRESULT SessionID (
        [out, retval] DWORD *pVal );
    [propput, id(DISPID_SAF_SESS_SESSIONID )] HRESULT SessionID (
        [in ] DWORD pVal );
    [propget, id(DISPID_SAF_SESS_SESSIONSTATE)] HRESULT SessionState(
        [out, retval] SessionStateEnum *pVal );
    [propput, id(DISPID_SAF_SESS_SESSIONSTATE)] HRESULT SessionState(
        [in ] SessionStateEnum pVal );
    [propget, id(DISPID_SAF_SESS_DOMAINNAME )] HRESULT DomainName (
        [out, retval] BSTR *pVal );
    [propput, id(DISPID_SAF_SESS_DOMAINNAME )] HRESULT DomainName (
        [in ] BSTR pVal );
    [propget, id(DISPID_SAF_SESS__USERNAME )] HRESULT UserName (
        [out, retval] BSTR *pVal );
    [propput, id(DISPID_SAF_SESS__USERNAME )] HRESULT UserName (
        [in ] BSTR pVal );
};

[
    object,
    uuid(F120A684-B926-447F-9DF4-C966CB785648),
    dual,
    nonextensible,
    helpstring("IRASrv Interface"),
    pointer_default(unique)
]

```

```

interface IRASrv : IDispatch{
    [id(1), helpstring("method GetNoviceUserInfo")] HRESULT GetNoviceUserInfo(
        [in,out] LPWSTR * szName);
    [id(2), helpstring("method GetSessionInfo")] HRESULT GetSessionInfo(
        [in,out] SAFEARRAY(BSTR) * UserNames, [in,out] INT * Count);
};

[
    uuid(833E4010-AFF7-4AC3-AAC2-9F24C1457BCE),
    helpstring("PCHService Class")
]
coclass PCHService
{
    [default] interface IPCHService;
}

[
    uuid(3C3A70A7-A468-49B9-8ADA-28E11FCCAD5D),
    helpstring("RASrv Class")
]
coclass RASrv
{
    [default] interface IRASrv;
};

```

8 Appendix C: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

The terms "earlier" and "later", when used with a product version, refer to either all preceding versions or all subsequent versions, respectively. The term "through" refers to the inclusive range of versions. Applicable Microsoft products are listed chronologically in this section.

Windows Client

- Windows XP operating system
- Windows Vista operating system
- Windows 7 operating system
- Windows 8 operating system
- Windows 8.1 operating system
- Windows 10 operating system
- Windows 11 operating system

Windows Server

- Windows Server 2003 operating system
- Windows Server 2008 operating system
- Windows Server 2008 R2 operating system
- Windows Server 2012 operating system
- Windows Server 2012 R2 operating system
- Windows Server 2016 operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 1.3](#): Windows implements **expert** authentication using the Offer **Remote Assistance** Helpers Group. **DCOM** will only allow members of this group to execute the Remote Assistance Initiation Protocol methods.

[<2> Section 1.7](#):

- Windows XP and Windows Server 2003 do not implement IRASrv. A **novice** running one of these versions of Windows cannot be initiated through IRASrv.
- Windows XP and Windows Server 2003 implement IPCHService. A Windows XP or Windows Server 2003 novice can be initiated only through IPCHService.
- A Windows Vista and later and Windows Server 2008 and later expert first attempts to call IRASrv, and if that fails, the expert then attempts to call the IPCHService interface.
- A Windows XP or Windows Server 2003 expert will only attempt to call IPCHService and will not be able to connect to a novice running Windows Vista and later or Windows Server 2008 and later.

<3> [Section 2.2.1](#): This form of the **Remote Assistance Connection** String is obtained from Remote Assistance running on a Windows XP or Windows Server 2003 novice.

<4> [Section 2.2.2](#): This form of the Remote Assistance Connection String is not available on Windows XP and Windows Server 2003.

<5> [Section 2.2.2](#): The KH2 attribute is not implemented in Windows XP or later.

<6> [Section 3.1.4.1](#): This interface and its called interfaces, IPCHCollection and ISAFSession, are only called by Windows XP and Windows Server 2003.

<7> [Section 3.1.4.1](#): Opnums 3 and 4 are used locally by Windows, never remotely.

<8> [Section 3.1.4.1.1](#): Access denied error is returned only for Windows XP operating system Service Pack 1 (SP1) and the initial release of Windows Server 2003.

<9> [Section 3.1.4.1.2](#): Access denied error is returned only for Windows XP SP1 and the initial release of Windows Server 2003.

<10> [Section 3.1.4.1.2.1](#): Gaps in the **opnum** numbering sequence apply to Windows as follows.

Opnum	Description
3-4	Only used locally by Windows, never remotely.

<11> [Section 3.1.4.1.2.2](#): Gaps in the opnum numbering sequence apply to Windows as follows.

Opnum	Description
3-4	Only used locally by Windows, never remotely.

<12> [Section 3.3.4.1](#): This interface is not implemented in Windows XP and Windows Server 2003.

<13> [Section 3.3.4.1](#): Gaps in the opnum numbering sequence apply to Windows as follows.

Opnum	Description
3-4	Only used locally by Windows, never remotely.

<14> [Section 5.1](#): For DCOM authentication/authorization in Windows Vista and later and Windows Server 2008 and later, the following are used:

- Authentication Service - RPC_C_AUTHN_WINNT
- Authorization Service - RPC_C_AUTHZ_DEFAULT
- Authentication Level - RPC_C_AUTHN_LEVEL_PKT_PRIVACY
- Impersonation Level - RPC_C_IMP_LEVEL_IMPERSONATE

For DCOM authentication/authorization in Windows XP and Windows Server 2003, all defaults are used.

[<15> Section 6](#): In Windows, the XML file has the extension MSRCIncident.

[<16> Section 6](#): This version is specific to Windows XP and Windows Server 2003.

[<17> Section 6](#): A password can be used in Windows XP and Windows Server 2003. For security reasons it is advisable to protect the Remote Assistance invitation file using a password.

In Windows XP and Windows Server 2003, when a password is used, it is encrypted using PROV_RSA_FULL predefined Cryptographic provider with MD5 hashing and CALG_RC4, the RC4 stream encryption algorithm. More details are in the Cryptography Reference [\[MSDN-CRYPTO\]](#).

The password is translated into a **Unicode string** before it is hashed. The **passStub** is a 14 character Unicode string. Also, the **passStub** is prefixed with a four-byte integer that contains the number of bytes in the following data string, as defined in BSTR data type ([\[MS-DTYP\]](#) section 2.2.5), when performing the encryption.

[<18> Section 6](#): This diagram applies to Windows XP and Windows Server 2003 implementations only.

[<19> Section 6](#): This type is specific to Windows Vista, Windows Server 2008, Windows 7, Windows Server 2008 R2 operating system, Windows 8, and Windows Server 2012.

[<20> Section 6](#): In Windows Vista, Windows Server 2008, Windows 7, Windows Server 2008 R2, Windows 8, and Windows Server 2012, the password is mandatory and it is encrypted using the MS_ENH_RSA_AES_PROV predefined Cryptographic provider with CALG_SHA hashing and the CALG_AES_128 encryption algorithm. The cipher mode used can be CRYPT_MODE_CBC and the password used during the encryption process is a Unicode string.

[<21> Section 6](#): This attribute can be present for Windows XP and Windows Server 2003 compatibility.

[<22> Section 6](#): This diagram applies to the following Windows implementations: Windows Vista, Windows Server 2008, Windows 7, Windows Server 2008 R2, Windows 8, and Windows Server 2012.

9 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
8 Appendix C: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

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—

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